```
!pip install sentence-transformers
```

```
Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.12/dist-packages (5.3.0)
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Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3
```

```
sentences = [
    # Cricket (3)
    "The spinner deceived the batsman with a sharp turning delivery",
    "Under pressure, the captain rotated the bowlers strategically",
    "A last over boundary sealed the victory for the chasing team",

    # Cooking (3)
    "The chef caramelized onions slowly to enhance the flavor",
    "A pinch of cinnamon completely changed the taste of the dish",
    "Let the dough rest so the bread becomes soft and airy",

    # Cybersecurity (4)
    "Multi-factor authentication adds an extra layer of account security",
    "Hackers often exploit weak network configurations to gain access",
    "Regular software updates help prevent security vulnerabilities",
    "Encrypted communication keeps sensitive information safe from attackers"
]
```

```
from sentence_transformers import SentenceTransformer

model = SentenceTransformer('all-MiniLM-L6-v2')

embeddings = model.encode(sentences)

print(embeddings.shape)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
modules.json: 100%  349/349 [00:00<00:00, 22.0kB/s]
config_sentence_transformers.json: 100%  116/116 [00:00<00:00, 4.93kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
README.md: 10.5k/? [00:00<00:00, 240kB/s]
sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 836B/s]
config.json: 100%  612/612 [00:00<00:00, 14.7kB/s]
model.safetensors: 100%  90.9M/90.9M [00:02<00:00, 209MB/s]
Loading weights: 100%  103/103 [00:00<00:00, 176.23it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100%  350/350 [00:00<00:00, 5.00kB/s]
vocab.txt: 232k/? [00:00<00:00, 3.03MB/s]
tokenizer.json: 466k/? [00:00<00:00, 6.08MB/s]
special_tokens_map.json: 100%  112/112 [00:00<00:00, 7.88kB/s]
config.json: 100%  190/190 [00:00<00:00, 3.67kB/s]
(10, 384)
```

```
from sklearn.metrics.pairwise import cosine_similarity

similarity_matrix = cosine_similarity(embeddings)
```

```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10,8))
sns.heatmap(similarity_matrix, annot=True, fmt=".2f")
plt.title("Sentence Similarity Heatmap")
plt.show()
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files embeddings_demo.ipynb

embeddings_demo.ipynb(n/a) - 27[illegible]8192 bytes, last modified: 3/20/2026 - 100% done

Saving embeddings_demo.ipynb to embeddings_demo.ipynb

```
import os
os.listdir()
```

```
['.config', 'embeddings_demo.ipynb', 'sample_data']
```

```
import nbformat

with open("embeddings_demo.ipynb", "r", encoding="utf-8") as f:
    nb = nbformat.read(f, as_version=4)

# Remove problematic widget metadata
if "widgets" in nb.metadata:
    del nb.metadata["widgets"]

with open("embeddings_demo_clean.ipynb", "w", encoding="utf-8") as f:
    nbformat.write(nb, f)

print("Clean notebook created!")
```

```
Clean notebook created!
```